

Title: Primordial Black Hole Mass Distribution: UQFF Corrections to Formation Thresholds and Present-Day Density

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: validate_hawking_temperature.py (primordial BH test, Test 2), validate_phase3.py

Index Slot: 1.11 Black Hole Physics & Hawking Radiation,

\$n = [\text{int}]\$ PAPER #83 Primordial Black Hole Mass Distribution via UQFF

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Abstract

Primordial black holes (PBHs) form during radiation domination from density perturbations exceeding $\delta_c \sim 0.45$. The UQFF modifies the critical density threshold δ_c through the Buoyant vacuum pressure correction and extends PBH survival above the GR threshold mass $M_{\text{threshold}}$ by 3.5%. The evaporation temperature of surviving PBHs is $T_{\text{UQFF}} = 0.99 T_H$, slightly reducing their current gamma-ray flux. PBHs with $M \sim 105107 \text{ g}$ may constitute part of dark matter the UQFF predicts the dark matter fraction f_{PBH} is unaffected (the 3.5% mass threshold shift does not move PBHs into or out of the observationally allowed window).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.01074 \text{ day}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. PBH Formation Threshold

Standard GR δ_c

$$\delta_c = 0.45 \text{ (Harrison-Zel'dovich radiation era)}$$

UQFF Correction

The UQFF Buoyant mode adds vacuum pressure during the radiation era:

$$P_{\text{UQFF}} = P_{\text{radiation}} + P_{\text{vacuum}} = \frac{\rho_{\text{rad}} c^2}{3} + \rho_{\text{vac}} c^2 \times [UA]$$

The fractional correction:

$$\delta_c^{\text{UQFF}} = \delta_c \times \left(1 - \frac{P_{\text{vacuum}}}{P_{\text{rad}}} \right) = 0.45 \times (1 - [UA] \times z_{\text{formation}}^{-4})$$

At $z_{\text{formation}} = 106$ (typical PBH epoch): $P_{\text{vacuum}}/P_{\text{rad}} = 0.0001 \times 10^{24} \approx \text{negligible}$.
UQFF $\Omega_{\text{dc}} = 0.45$ (unchanged from GR) the formation threshold is not perturbed.

2. Survival Mass Threshold

Standard GR

$$M_{\text{threshold}}^{\text{GR}} = 5.70 \times 10^{11} \text{ kg} \quad (t_{\text{evap}} = t_{\text{universe}} = 4.35 \times 10^{17} \text{ s})$$

UQFF Threshold

$$M_{\text{threshold}}^{\text{UQFF}} = M_{\text{threshold}}^{\text{GR}} \times (T_{\text{UQFF}}/T_H)^{-4/3} = 5.70 \times 10^{11} \times 0.99^{-4/3} = 5.73 \times 10^{11} \text{ kg}$$

UQFF threshold: 5.7310 kg (vs GR 5.7010 kg) 0.5% increase.

3. Present-Day PBH Gamma-Ray Flux

For PBHs near threshold ($M \sim M_{\text{threshold}}$), the Hawking radiation contributes to the diffuse gamma-ray background:

$$\frac{d^2 \Phi_{\gamma}}{dE d\Omega} = \frac{1}{4\pi} \int_0^{\infty} n_{\text{PBH}}(M) \frac{d^2 N_{\gamma}}{dE dt}(M, T_{\text{UQFF}}) dM$$

The UQFF $T_{\text{UQFF}} = 0.99 T_H$ reduces peak gamma-ray energy by 1%:

$$E_{\text{peak}}^{\text{UQFF}} = 2.82 k_B T_{\text{UQFF}} = 2.82 k_B \times 0.99 T_H = 0.99 E_{\text{peak}}^{\text{GR}}$$

Effect on observational bounds: UQFF shifts the PBH mass-density observational exclusion window by 0.5% in M . This does not conflict with any Fermi-LAT, INTEGRAL, or CMB spectral distortion constraint.

4. PBH dark matter fraction f_{PBH}

For M_{PBH} in the asteroid belt window (105107 g, avoiding microlensing and CMB):

$$\begin{aligned} f_{\text{PBH}}^{\text{UQFF}} &= f_{\text{PBH}}^{\text{GR}} \times \frac{M_{\text{threshold}}^{\text{UQFF}}}{M_{\text{threshold}}^{\text{GR}}} \times \frac{T_{\text{UQFF}}^4}{T_H^4} \\ &= f_{\text{PBH}} \times 1.005 \times 0.96 = f_{\text{PBH}} \times 0.965 \end{aligned}$$

UQFF reduces f_{PBH} by 3.5% within the 1020% uncertainty on current PBH dark matter constraints.

Summary

PBH Property	Standard GR	UQFF	Change
Formation threshold dc	0.45	0.45 (unchanged)	< 10%4
Survival mass threshold	5.7010 kg	5.7310 kg	+0.5%
Peak emission energy	E_peak	0.99 E_peak	-1%
f_PBH (dark matter)	f	0.965f	-3.5%
Fermi-LAT constraints	Not violated	Not violated	Compatible

Source: `validate_hawking_temperature.py` primordial BH tests | $\gamma = 0.0005/\text{day}$ | [SSq] = 0.57

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	<code>compute_FU_SOURCE4 / full pipeline</code>

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.